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How Much Training Data Has a Genome Seen?

An Algorithmic-Information View of Evolutionary Optimization Depth

At the Cellular Coarse-Graining Scale

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Abstract

Why can a child learn many concepts from a few examples, while artificial neural networks need enormous corpora? One answer is that the child is not learning from scratch: the human nervous system instantiates priors shaped by deep evolutionary filtration. We estimate the order of magnitude of that filtration at the cellular coarse-graining and compare it to LLM pretraining on two unit-respecting axes:

- **Axis 1 — Stored object.** A diploid human genome is $\sim 10^{10}$ bits; the weights of a frontier LLM (Llama 3.1 405B) are $\sim 10^{13}$ bits. The genome is $\sim 10^{3}\times$ *smaller*.
- **Axis 2 — Exposure / evaluation stream.** A frontier LLM is pretrained on $\sim 10^{13}$ token exposures; the biospheric cell-level stream that filters extant genomes is $\sim 10^{38}$–10^{40} exposure / evaluation events. Biological filtration depth is $\sim 10^{25}\times$ *larger*, as an event-count comparison (not an information-equivalence claim).

The cross-comparison — genome bits versus LLM corpus bits — is not made: stored objects and exposure streams are different quantities. The mechanisms also differ: LLMs receive dense differentiable error over token streams; evolution receives sparse, delayed, population-level feedback through survival and reproduction. The key result is not that DNA stores ancestral data, but that genomes are compact survivor programs produced by a vastly deeper but much sparser write-back process than gradient descent. This evolutionary prior helps explain why biological learners begin not from scratch, but from deep evolutionary structure.

Keywords: algorithmic information theory, Kolmogorov complexity, evolution, coarse-graining, genome, large language models, write-back, persistence, algorithmic agent

Classification: MSC 68Q30, 92D15 / ACM I.2.0, J.3

Contents

1 Introduction

Frontier large language models (LLMs) are pretrained on token corpora of order 10^{12}–10^{13} events.[1,2] A human child, by contrast, can learn many concepts from a handful of examples. The simplest reading of this gap is that the child is not learning from scratch: by the time individual learning begins, the human nervous system is the deployment of a hereditary program shaped by a much longer training process. The question this paper addresses is: *how much longer*, and on what unit. The naive cross-species answer — "billions of years" — is correct in spirit but useless for comparison. The legitimate move is to count evaluation events at a shared coarse-graining and ask, in Lloyd's spirit,[3] how much optimization exposure has shaped the patterns that persist today.

Two foundational moves anchor the analysis and should be stated up front.

Move 1 — Coarse-graining is constitutive of the question. Microscopically, every organism is a vast physical computer: a single cell runs $\sim 10^{13}$–10^{17} molecular operations over its lifetime; a human lifetime is $\sim 10^{26}$–10^{30} cellular operations and $\sim 10^{20}$ neural events. None of these elementary counts is the right unit for "biological learning." Following the Wolpert–Korbel-style account of computation between dynamical systems formalized in WP0049,[4] we fix a coarse-graining $C_s : X(t) \mapsto P_s(t)$ and ask only about the dynamics of patterns visible at scale s (molecule, cell, organism, lineage, population, culture). Microscopic computation is abstracted away. The question of evolutionary training is well-posed at the cellular, organismic, and lineage scales, not at the molecular one.

Move 2 — Training is pattern–world interaction at the chosen scale. A *training event* at scale s is an interaction

$$e_k = (P_s(t),\, E_s(t),\, \Delta t) \tag{1}$$

between the pertinent coarse-grained pattern P_s and the rest of the world $E_s = W_s \setminus P_s$ that changes the future generator of P_s. This is the operational definition of "training" inside the algorithmic-agent framework.[5,6] It makes the comparison to LLM pretraining precise: each token event is a pattern–corpus interaction that updates parameters θ in a fixed graph; each cell–environment interaction is the biological counterpart, but the update is sparse, scalar, and end-of-lifetime — it acts through the reproductive channel at the lineage level, mediated by the write-back vector $\boldsymbol{\lambda} = (\lambda_B, \lambda_F, \lambda_P)$ of WP0058.[7]

Headline result, on two separate axes. Under these two moves, two unit-respecting comparisons are available, and they should not be crossed:

1. **Stored object (the compressed survivor).** It takes $\sim 10^{10}$ bits to specify a diploid human genome and $\sim 10^{13}$ bits to specify the weights of a frontier LLM (Llama 3.1 405B at FP32). The genome is $\sim 10^{3}\times$ *smaller.*

2. **Training input (the upstream filtration).** An LLM pretraining corpus is $\sim 10^{14}$–10^{15} bits; the cell-level biospheric exposure stream is $\sim 10^{38}$–10^{40} events (toy-coded conservatively at one binary-valued evaluation per event). Evolutionary filtration input is $\sim 10^{23}$–$10^{26}\times$ *larger.*

The genome is a *compact hereditary description relative to the ensemble of organism-generating programs explored and filtered by evolution* — not a serial training history experienced by any one lineage, and not a stored summary of 10^{40} events. The mammalian sub-lineage N_eG (where

N_e is the effective population-genetic size, not a count of organisms that lived) is $\sim 10^{12}$–10^{13}, comparable in order of magnitude to frontier LLM token counts but on the input axis.

Where the LLM parallel breaks. We frame the comparison as one of *exposure depth*, not architectural identity. Evolution is program search (it can rewire the graph), end-of-lifetime sparse reward, population-mediated, and embodied. LLM pretraining is parameter search in a designer-fixed graph, dense per-token gradient, single-instance, and disembodied. The shared unit is not mechanism but *number of evaluation events absorbed by the optimizer at the chosen coarse-graining scale.*

Structure. Section 2 establishes the coarse-graining and training-event definitions. Section 3 gives a multi-scale event taxonomy distinguishing the slow reproductive channel from faster within-lifetime channels. Section 4 works the numbers. Section 5 states the AIT punchline: DNA as compressed survivor program. Section 6 hedges the LLM parallel. Section 8 concludes.

2 Foundational moves: coarse-graining and the training event

2.1 The coarse-graining choice

Let $X(t)$ denote the microscopic physical state of the world at time t. A coarse-graining is a (computable) projection

$$C_s : X(t) \longmapsto P_s(t), \tag{2}$$

extracting a pattern P_s at scale s. The choice of s is not innocent. It commits to (i) the variables that are explanatorily or causally relevant, (ii) the dynamics that are predictable at that level, and (iii) the update rule that the system is taken to obey. WP0049[4] formalizes this in the Wolpert–Korbel framework: a system A is said to compute another system B when there exist coarse-grainings such that $C(A) \cong C(B)$ as dynamical systems. Equivalently, computation — and learning, as a special case — is a relation between models of dynamical systems at specified scales, not a primitive property of microdynamics.

For the present question, the natural ladder is:

1. **Cellular pattern** — membrane, metabolism, genome, regulatory network.
2. **Organism pattern** — developmental body plan, nervous system, physiology, behavior.
3. **Lineage pattern** — population distribution over hereditary programs.
4. **Agent pattern** — model-building, objective-guided, action-generating system.[5,6]
5. **Human pattern** — mammalian, primate, hominin, cultural, linguistic agent.[8]

We do not need to simulate quarks to study mammalian learning, just as we do not need transistor-level dynamics to describe LLM pretraining. The relevant question is: *at what scale does the pattern maintain identity, interact with the world, receive feedback, and update?* Below that scale, computation is abstracted away. At and above it, training is visible.

2.2 Training as pattern–world interaction: a three-level taxonomy

At scale s, split the world into the pattern and its environment, $W_s(t) = P_s(t) \cup E_s(t)$. The single notion of "training event" is too coarse: it conflates four distinct things — data exposure, evaluation, write-back, and physical compute. The first half of this section disentangles them.

Definition 1 (Three-level training taxonomy at scale s)

An **event** at scale s is an interaction between the pattern P_s and its environment E_s that carries *at least one bit* (typically many more) of information about the world. The three nested kinds of event we count are:

Exposure event ξ — the pattern samples its environment, transducing at least one bit of environmental state. (Biological: cellular sensing, metabolism, immune encounter, behavior, sensory episode. ML: a token or context exposure.)

Evaluation episode η — an exposure that receives a persistence- or reproduction-relevant score, contributing at least one bit of fitness-relevant feedback. (Biological: cell division, survival, mating, offspring viability. ML: loss computed on a batch.)

Write-back event ω — an evaluation whose result modifies the future generator of P_s. (Biological: allele or genome-frequency shift, epigenetic inheritance, horizontal gene transfer, cultural transmission. ML: parameter update.)

The thesis of this paper is then sharper and harder to attack:

Thesis (Evolutionary training as a chain)

Evolutionary training of a hereditary pattern is the *chain* $\xi \to \eta \to \omega$: exposure episodes are scored by persistence and reproduction, and a thin filtered residue is written back into the future distribution of hereditary generators. Each level is part of the training process. An exposure that does not directly trigger a write-back is not "noise" — it is part of the upstream depth that determines what gets filtered. The cellular exposure budget therefore bounds the *filtration depth* of the process, while the (much smaller) write-back rate bounds what gets retained in H_g.

Why exposure without write-back still counts as part of training. An exposure that does not immediately rewrite H_g can still shape the chain: it modifies the runtime state, the behavior, the survival probability, and (collectively, across many similar agents) the population-level fitness landscape that selection eventually integrates. The information that gets written back is what it is *because* the upstream exposure depth is what it is. Treating only write-back events as "real learning" would discard the upstream variable that determines what is filtered. The right way to read the three levels is as stages of one process, not a wall between training and non-training.

Data seen by the organism is not equal to data written into the genome. The vast majority of biological exposures terminate at the evaluation stage; only a thin filtered residue is written back. The cellular exposure budget therefore reports the depth of the filtration, not the number of bits stored in any genome.

Table 1: Three kinds of learning, by which pattern is updated.

Type	Pattern updated	Heritable?
Runtime learning	Cell state, immune memory, neural weights, physiological regulation	Usually no
Hereditary learning	Population distribution over hereditary programs H_g	Yes (Darwinian)
Cultural / niche learning	Externalized symbolic and material structure	Partly; not DNA-only

Three kinds of learning, by where they write. Within the WP0058 framework,[7] all three are special cases of persistence dynamics. The write-back vector $\boldsymbol{\lambda} = (\lambda_B, \lambda_F, \lambda_P)$ controls how much within-lifetime learning enters the inherited description: $\boldsymbol{\lambda} \to \mathbf{0}$ is strict Darwinian inheritance (this paper's main object of study), large $\boldsymbol{\lambda}$ approaches Lamarckian transmission, and modern AI training and fine-tuning sit at the upper corner. The seven-stage substrate arc of WP0110[8] unfolds along $\boldsymbol{\lambda}$ as new substrates accumulate.

Remark 2 (Scope: this paper studies the Darwinian channel)

The quantitative estimates that follow focus on the Darwinian-dominated regime ($\boldsymbol{\lambda} \to \mathbf{0}$): germline inheritance with sparse, end-of-life, scalar reproductive feedback. We are aware that Lamarckian-style write-back is real and non-trivial: epigenetic inheritance (DNA methylation, histone marks, small-RNA pathways), horizontal gene transfer, the holobiont microbiome, niche construction, cultural transmission, and emerging human–AI symbiosis all sit at moderate-to-large $\boldsymbol{\lambda}$. These are treated in WP0058 (the continuum)[7] and WP0110 (the substrate arc).[8] They do not change the order-of-magnitude story for the cell-level biospheric exposure budget reported here, but they raise the effective write-back rate above the Worden / Kimura floor for any specific lineage in proportion to the substrate stack it has accumulated.

Remark 1 (Why coarse-graining is constitutive

The argument for coarse-graining is *not* that the molecular trajectory is incompressible. On the contrary, the joint trajectory of 10^{26}–10^{30} chemical events in an organism's lifetime is highly compressible by the laws of physics — that is what physics *is*. The point is rather that, for the computation we care about, the precise micro-state is irrelevant. What matters is "this protein, in this place, in this configuration," not the exact positions and momenta of the atoms that realize it. Coarse-graining works because, at the chosen scale, an enormous equivalence class of micro-states maps to the same coarse pattern, and the coarse pattern's dynamics close on themselves (approximately) under the rules of that scale. Choosing the scale is choosing the equivalence relation under which the question of learning becomes well-posed.

3 Multi-scale event taxonomy

3.1 Dimensional rosetta: what is and is not commensurable

Theoretical stance: training data ≡ interactions with environment. The unifying conceptual move that licenses the comparison is this: in both biology and ML, "training data" should be read as the agent / pattern's *information flux with its environment*. For an LLM, the environment is a curated text corpus, and the interactions are token / context exposures. For a biological pattern, the environment is the physical / chemical / ecological world, and the interactions are cell-environment, organism-environment, and lineage-environment events. The two situations differ in substrate, in update rule, and in the shape of the training signal — but they share the role of being the pattern's flux with its environment at the chosen coarse-graining. The rest of this paper compares biology and ML *within* that shared role.

Five dimensions, never crossed. Before counting, fix the units. Five dimensions of the evolution / ML comparison must be kept apart:

Table 2: Dimensional rosetta for the evolution / ML comparison. Quantities in the same row are commensurable (with caveats); quantities in different rows are not.

Dimension	Biology	LLM
Stored object	Genome (bits) / cellular machinery	Model weights / training corpus (bits)
Exposure stream	Cell-environment, organism-environment interactions	Token / context exposures
Evaluation signal	Survival, reproduction, fitness differentials	Prediction loss / cross-entropy
Write-back channel	Allele-frequency change, epigenetics, HGT, cultural transmission	Gradient parameter update
Physical compute	Cellular and molecular operations	FLOPs

The rest of this paper compares quantities *within rows*: events to events, bits to bits, FLOPs to FLOPs. Cross-row comparisons are avoided, and where they appear (e.g. Lloyd-style accounting of biospheric molecular ops next to LLM training FLOPs) they are flagged as loose.

3.2 Per-scale event budgets

Biology runs an information-processing hierarchy with very different update rules at each level. Only the slowest (reproductive) signal updates the heritable program H_g directly; faster signals shape each evaluation. Table 3 summarizes.

Table 3: Multi-scale event taxonomy. Counts are reported at the appropriate natural unit for each scale; the last column states that unit explicitly. Order-of-magnitude estimates; uncertainties are large by construction.

Scale	Event	Count	Per	Updates / Mechanism
Molecular	Enzymatic / signaling reaction	10^{13}–10^{17}	cell lifetime	Transient cellular state
Cellular	Cell division	10^{11}–10^{13}	human-body lifetime	Soma; genome via mutation
Sensory / neural	Spike, synaptic event	10^{19}–10^{23}	human lifetime	Synaptic weights, immune memory
Behavioral	Sensorimotor episode	10^{10}–10^{12}	individual lifetime	Behavior, culture
Generational	Reproduction (or failure)	1	individual	Allele frequencies in lineage
Phylogenetic	Speciation, extinction	$\sim 10^{-6}$	yr per lineage	Clade-level repertoire
Biospheric	Cell-level production / evaluation	10^{38}–10^{40}	Earth history	Filtration depth (this paper)

Four observations follow.

(a) Reproduction is the reinforcement-learning reward. Of all these events, only the generational scale writes to H_g. Failure to reproduce is the loss signal. It is sparse (one per lifetime), delayed (after the whole embodied evaluation has run), and unidimensional (essentially scalar), but it is the only channel by which lifetime experience couples back into the inherited program.[7]

(b) Lifetime is computation, not a test set. The framing of within-lifetime activity as "test data" that is run against a fixed program and then discarded is too thin. A lifetime is itself the *evaluation of the model against the world*, and the running of that model is precisely the computation that generates the error signals that the system uses — across cells, organisms, and lineages — to decide what gets culled and what gets passed along. The reproductive signal at end-of-life is the integrated output of this computation, not a label attached to a passive datum. Coarse-graining lets us count generational evaluations as the unit at which heritable updates occur, but it should not be mistaken for the claim that the runtime is informationally inert.

(c) Lifetime computation is collective. An additional point that the test-set framing erases: agents with similar pattern — formally, agents sharing high mutual algorithmic information (MAI) at the chosen coarse-graining — interact and *compute together* during their lifetimes. Cells in a tissue, organisms in a population, conspecifics in a niche, and members of a culture jointly run a computation whose outputs include error signals that feed both within-lifetime adaptation (immune, neural, behavioral) and the slow reproductive channel that updates H_g. This collective computation is consistent with the multilevel-learning view[9] and is treated formally in our agent-soup work (WP0056). Persistent patterns persist in part *because* similar patterns can interact to extract information that no single instance could extract alone — a point reinforced by Frank's information-theoretic reading of selection[10] and by the WP0058 write-back continuum.[7]

(d) Somatic and neural learning are real but ephemeral. Immune systems, brains, and epigenetic state acquire information within a lifetime — sometimes a great deal of it.[11] But this somatic learning dies with the individual unless it is captured by the slow channel (selection on neural priors, the Baldwin effect, or cultural transmission in social species).[8] Lamarckian write-back to DNA is, in mammals, mostly disallowed.

(e) The genome integrates events that no ancestor directly experienced. A modern mammalian genome carries optimization from $\sim 10^{40}$ ancestral cellular evaluations, including 2–3 Gyr of prokaryotic and protist history that predates the mammal pattern itself. The mammal-specific layer is a thin recent refinement on this stack.

4 Quantitative estimates

4.1 Biospheric cellular exposure: $\sim 10^{38}$–10^{40} events

What this number is, and is not. The estimate that follows is a *biospheric parallel exposure budget* — the order-of-magnitude count of cell-level production / evaluation events the global biosphere has generated since early life. Earth history is not a single panmictic optimizer; horizontal gene transfer, niche structure, and ecological partitioning mean the budget is parallel across a vast distributed search rather than a serial training history experienced by any one lineage. Extant genomes are products of *lineages embedded in* this global search, not direct recipients of all of its events.

Whitman, Coleman and Wiebe estimated present-day prokaryotic production at $P_0 \approx 1.7 \times 10^{30}$

cells/yr and a global standing population of 4–6 $\times 10^{30}$ cells.[12] Moody et al.[13] reconstruct the last universal common ancestor at ~ 4.2 Ga; conservative early-life evidence reaches back to ~ 3.7 Ga (contested).[14,15] An upper-bound anchor over $T \in [3.7, 4.2]$ Gyr is

$$E_{\text{cell}}^{\text{upper}} \;=\; P_0\, T \;\approx\; 6.3 \times 10^{39} \text{ to } 7.1 \times 10^{39}. \tag{3}$$

Discounting for lower early-Earth productivity (effective window ~ 2 Gyr at 0.1–1 of modern rate) yields a defensible range

$$\boxed{E_{\text{cell}} \;\sim\; 10^{38}\text{–}10^{40} \quad \text{cell-level production / evaluation events.}} \tag{4}$$

This is the depth of the global parallel filtration from which extant lineages descend. Compared to LLM token counts ($\sim 10^{12}$–10^{13}),[1,2] this is 25–28 orders of magnitude greater exposure — with the qualification that biological events and tokens are not literally the same kind of object (cf. the rosetta of §3.1).

4.2 Mammalian N_eG as a secondary sanity check

A secondary, narrower comparison: the lineage-scale hereditary search proxy N_eG for mammals, with mammalian history $T_m \in [200, 225]$ Myr,[16,17] generation length $g \in [1, 5]$ yr,[18] and the comparative-genomics estimate of long-term mammalian effective population size $N_e \approx 6.1 \times 10^4$,[19] gives $G_m = T_m/g \approx 4 \times 10^7$ to 2.25×10^8 generations and

$$N_e\, G_m \;\approx\; 2.4 \times 10^{12} \text{ to } 1.4 \times 10^{13}. \tag{5}$$

What N_eG is and is not. The effective population size N_e is a population-genetic quantity related to genetic drift, coalescence, and selection efficacy — *not* the number of organisms that lived, and not an event counter.[19] The product N_eG is therefore best read as a lineage-scale hereditary search proxy: *at the effective-population coarse-graining*, the mammalian lineage's search depth falls near frontier LLM token counts in order of magnitude. This is a sanity check on the cellular headline of §4.1, not a replacement for it. The dramatic comparison remains at the cellular scale.

4.3 Within-lifetime physical compute, compared to FLOPs (not tokens)

A single human lifetime runs an enormous physical computation, but this is *not* commensurable with LLM tokens and must not be put in the same column. The legitimate comparison is against LLM training FLOPs.

LLM training compute. A crude dense-transformer estimate for Llama 3.1 405B is

$$C_{\text{LLM}} \;\approx\; 6\, N\, D \;=\; 6 \times 4.05 \times 10^{11} \times 1.5 \times 10^{13} \;\approx\; 3.6 \times 10^{25} \text{ FLOPs}, \tag{6}$$

in line with Chinchilla's reported 5.76×10^{23} FLOPs at the smaller (70B / 1.4T) scale.[1,2]

One human lifetime. For a single human lifetime ($\sim 2.5 \times 10^9$ s, $\sim 3 \times 10^{13}$ cells):[20,21]

- **Neural spike / synaptic events:** $\sim 10^{20}$–10^{23} — *below* C_{LLM}.
- **Molecular operations:** $\sim 10^{26}$–10^{30} — $\sim$ 1–5 OOM *above* C_{LLM}.
- **Cellular turnover events:** $\sim 10^{16}$.[22]

Molecular operations are not FLOPs, and the cross-substrate comparison is loose; even so, the order-of-magnitude reading is that one human lifetime's physical activity is in the same ballpark as a frontier LLM's training compute (above on molecular accounting, below on neural accounting). The dramatic comparison is *not* here.

Biosphere total. Aggregated over $\sim 10^{40}$ cell-lifetimes, total biological physical compute is, in Lloyd's spirit,[3]

$$\text{biosphere total} \;\sim\; 10^{53} \text{ molecular operations} \quad (10^{51}\text{–}10^{57}), \tag{7}$$

~ 65 OOM below Lloyd's cosmic 10^{120}, and ~ 25–30 OOM above C_{LLM}. Biological systems massively exceed LLMs in embodied, parallel, lifetime-scale physical activity. The robust comparison, however, is not "human lifetime compute versus LLM FLOPs" but *evolutionary exposure depth versus LLM token / loss exposure.*

4.4 Two comparison tables, kept separate

Table 4: **Event-count comparison.** Exposure / evaluation events at the indicated coarse-graining. Tokens and biological events are not literally the same kind of thing, but they share the role of "unit of exposure at the coarse-graining of interest."

Quantity	Value (events)	Relative to LLM tokens
LLM pretraining (Chinchilla)[1]	1.4×10^{12}	baseline
LLM pretraining (Llama 3.1)[2]	1.5×10^{13}	baseline
Mammal-lineage $N_e G$ (germline-level evaluation)	2.4×10^{12}–1.4×10^{13}	comparable
Cellular lineage evaluations	10^{38}–10^{40}	+25 to +28 OOM

Table 5: **Physical-compute comparison.** Reported in operations / FLOPs. Molecular operations are not FLOPs and the cross-substrate comparison is loose; we report it only to anchor the scale.

Quantity	Value	Notes
Chinchilla training compute[1]	5.76×10^{23} FLOPs	70B params, 1.4T tokens
Llama 3.1 family training[2]	~ 39M H100 GPU-hours	405B at 30.84M GPU-hours
One human lifetime, neural events	10^{20}–10^{23}	spike / synaptic events
One human lifetime, molecular ops	10^{26}–10^{30}	enzymatic / signaling reactions
Biosphere history, molecular ops	10^{51}–10^{57}	Lloyd-style accounting
Lloyd cosmic bound[3]	$\lesssim 10^{120}$	ops on $\sim 10^{90}$ bits

5 The AIT picture: DNA as compressed survivor program

5.1 What this paper does *not* claim

Before the bit-budget table, six claims must be explicitly disowned, because casual reading of the headline number invites all of them:

1. that DNA stores ancestral experiences as data;
2. that biological events are equivalent to tokens;
3. that 10^{40} events equals 10^{40} independent or informative bits;

4. that evolution performs gradient descent;
5. that current genomes are globally optimal;
6. that the full biospheric history is a single lineage-specific training set.

The rest of this section reports order-of-magnitude budgets under explicit conventions; the budgets do not assert any of (1)–(6). Selection is a lossy, sparse, population-mediated write-back channel.[10] The retained heritable information is far smaller than the upstream exposure; only the information that affects differential persistence and reproduction is fixed, and the per-generation fixation rate is bounded under restrictive assumptions to roughly order-one bits or less.[23]

5.2 Bit budgets, kept separate: storage vs. exposure

Table 6: **Stored-description sizes (apples to apples in bits).** For genomes, bases × 2 bits per base (4 nucleotides). For LLM weights, params × precision. All figures order-of-magnitude.

Stored object	**Bits**	**Notes**
Raw haploid human genome	$\sim 6 \times 10^9$	3×10^9 bases × 2 bits/base[24]
Diploid genome	$\sim 1.2 \times 10^{10}$	two parental copies; ignores epigenetics
Functionally constrained genome ($\sim 8.2\%$)	$\sim 10^9$	fraction under purifying selection[25]
Chinchilla 70B weights (FP32)	$\sim 2.2 \times 10^{12}$	7×10^{10} params ×32 bits
Llama 3.1 405B weights (FP16/BF16)	$\sim 6.5 \times 10^{12}$	4.05×10^{11} params ×16 bits[2]
Llama 3.1 405B weights (FP32-equivalent)	$\sim 1.3 \times 10^{13}$	4.05×10^{11} params ×32 bits
Chinchilla training corpus	$\sim 10^{13}$–10^{14}	$\sim 1.4 \times 10^{12}$ tokens[1]
Llama 3.1 training corpus	$\sim 10^{14}$–10^{15}	$\sim 1.5 \times 10^{13}$ tokens
One human lifetime, sensory exposure	$\sim 10^{16}$	retinal channel $\sim 10^7$ bit/s over ~ 70 yr

Table 7: **Exposure / evaluation budgets.** Event counts only. Each event is an interaction carrying at least one bit (typically many more) of information about the world (Def. 1). The retained heritable information after selection is much smaller and not estimated here.

Quantity	**Events**
LLM pretraining (Chinchilla[1] / Llama 3.1[2])	$\sim 10^{12}$–10^{13} token exposures
Mammalian N_eG (effective-population search depth)[1]	$\sim 10^{12}$–10^{13}
Cell-level biospheric exposure / evaluation stream	$\sim 10^{38}$–10^{40}
Biosphere total physical computation (ops, not bits)	$\sim 10^{51}$–10^{57}

On a toy binary coding (footnoted, not headlined). [2]

Two readings, on *the two correct axes*, sharpen the comparison. The wrong axis is genome bits versus LLM corpus bits; that is comparing a stored object to a training input and is not done in this paper.

Axis 1 — Stored object: LLM weights vs. genome. It takes $\sim 10^{13}$ bits to specify the weights of Llama 3.1 405B (at FP32) and $\sim 10^{10}$ bits to specify the diploid human genome (at 2 bits/base). The genome is $\sim 10^3\times$ *smaller* as a stored object than a frontier LLM, despite

[2] Purely illustrative: if one assigns a binary code to each survival/division opportunity, the cell-level exposure stream can be *visualized* as 10^{38}–10^{40} binary-valued trials. This is not an estimate of accumulated information, not a claim about heritable bits, and not commensurable with text-corpus bits. We do not use it elsewhere in the paper, where event counts are the primary unit.

being the survivor of a much deeper upstream filtration. The genome is a *more parsimonious* compressed survivor.

Axis 2 — Exposure / evaluation events: LLM tokens vs. biospheric stream. A frontier LLM is pretrained on $\sim 10^{13}$ token exposures. The cell-level biospheric exposure / evaluation stream is $\sim 10^{38}$–10^{40} events (Def. 1). At their respective coarse-grainings, biological exposure / evaluation events exceed LLM token exposures by $\sim 10^{25}$ orders of magnitude. This is an event-count comparison, not an information-equivalence claim. The retained heritable information is far smaller in both cases — selection (in the biological case) and gradient descent (in the LLM case) are lossy write-back channels.

Compression-ratio readings. Within each substrate, the survivor-to-input ratio is approximately: LLM weights / training corpus $\sim 10^{-1}$ to 10^{-2} (real, in commensurable bits); genome / cell-event budget $\sim 10^{-30}$ (heuristic — cf. Eq. (8) below). The two ratios are not directly comparable in their current form, but they make the qualitative point: **evolution produces a smaller compressed survivor from a vastly larger upstream search; LLMs produce a larger compressed survivor from a smaller upstream input**.

Event-to-description ratio. The ratio

$$\frac{|\pi|}{|\,\text{events}|} \sim \frac{10^{10}\ \text{bits genome}}{10^{40}\ \text{events}} = 10^{-30} \tag{8}$$

is *not* an AIT compression ratio — the denominator is event count, not the Kolmogorov description length of the world-stream. We name it an **event-to-hereditary-description ratio** and read it as a scale-separation heuristic: a finite hereditary program of order 10^{10} bits is the survivor of a search process of order 10^{38}–10^{40} events.

5.3 The survivor-program hypothesis

Hypothesis 1 (Genome as compressed survivor program)

At the cellular and lineage coarse-grainings, the genome π is not a stored archive of ancestral experience. It is a *compact hereditary description relative to the ensemble of organism-generating programs explored and filtered by evolution* — a survivor conditional on, not a summary of, the upstream exposure stream:

$$\pi \sim \text{Filter}_{\text{selection}}(D;\ \text{persistence/reproduction}),$$

with D the historical stream of organism–environment interactions. Selection is a very lossy write-back channel: only the information that affects differential persistence and reproduction is retained.[10] Under restrictive assumptions the per-generation fixation rate is bounded to roughly order-one bits or less,[23] though the exact bound depends on population structure, recombination, selection pressure, and what counts as fixed.[9,10]

This is the algorithmic-information reading of natural selection: a population accumulates information about the environment[10] via persistence-weighted filtration, and the resulting program is short *relative to the variety of programs selection has visited and rejected* (not necessarily short in absolute Kolmogorov sense). The closely-related computational analogue is Koza-style genetic programming[26] or Levin search[27] over a space of programs of bounded length — not parameter optimization in a fixed graph. The architecture is itself a target of selection; the genome encodes its own developmental and computational machinery.[7,8]

In the language of WP0077,[28] a mammal is a *persistent pattern* with high algorithmic probability in its native environment: a compact program located by deep evolutionary filtration in program space. Its persistence is not coincidence; it is the lawful consequence of being a survivor of that filtration.

Remark 3 (Vanchurin et al. on evolution as multilevel learning)

The framework here is fully compatible with the multilevel-learning view of Vanchurin, Wolf, Katsnelson and Koonin,[9] who treat biological evolution as learning across physically renormalizable levels. The present coarse-graining picture is the AIT face of the same idea: each level s has its own exposure, evaluation, and write-back events, and its own compressed survivor program.

6 Where the LLM parallel holds, and where it breaks

The comparison is intentionally an order-of-magnitude statement about optimization exposure, not an architectural identity. Table 8 summarizes.

Table 8: Evolutionary versus LLM training as search processes.

	Evolution (cellular / lineage scale)	**LLM pretraining**
Search space	Program structure + content	Parameter values in a fixed graph
Population	Many genomes per generation	One parameter vector
Update rule	Differential reproduction (sparse, scalar, end-of-life)	Per-token gradient (dense, vectorial)
Architecture	Itself a target of optimization	Fixed by designers
Substrate	Embodied; each evaluation runs an organism	Disembodied forward pass
Timescale	Generational / geological	Wall-clock during training
Event count	10^{38}–10^{40} cellular evaluations	10^{12}–10^{13} token events
Compressed survivor	Genome π ($\sim 10^{10}$ bits)	Weights θ ($\sim 10^{11}$ params)

Where it holds. Both processes are search procedures operating on a parameter / program set, scored by a loss-like signal, accumulating compressed structure that captures regularities of the environment. The natural shared unit is *number of evaluation events at a fixed coarse-graining.* By that unit, life's optimization budget exceeds frontier ML's by 25–28 OOM at the cellular scale; mammalian selection alone matches frontier LLM token counts.

Where it breaks.

- **Signal density.** LLM training provides dense, differentiable per-token gradients. Evolution provides a sparse, scalar, end-of-lifetime reproductive signal with no gradient information about which alleles caused the outcome.
- **Parallelism.** Evolution is massively parallel and population-mediated; LLM training is sequential mini-batch SGD on a single parameter set.
- **Architecture.** A genome specifies a developmental program, not a learned function. The "model" is the resulting organism plus its substrate; the "parameters" are distributed across DNA, protein structure, cellular machinery, and ecological context.
- **Update target.** Most lifetime data ingested by an organism never updates π. Only the small fraction captured by reproductive differentials propagates.

- **No global N_e.** Earth history is not panmictic. Horizontal gene transfer, niche structure, and ecological partitioning mean the cell-level event count is not an effective population size in the strict population-genetic sense.

The correct framing is therefore informational and algorithmic, not literal. On the *stored-object* axis, selection converges to a hereditary program π of size $\sim 10^{10}$ bits; LLM training converges to a parameter vector of $\sim 10^{13}$ bits. The genome is $\sim 10^3\times$ *smaller*. On the *training-input* axis, selection filters through $\sim 10^{38}$–10^{40} cell-level events; LLM training reads $\sim 10^{14}$–10^{15} corpus bits. Evolutionary input is $\sim 10^{24}\times$ *larger*. The two stored objects differ by ~ 3 OOM (LLM larger), the two upstream inputs differ by ~ 24 OOM (evolution larger), and the architectures differ qualitatively.

7 Discussion: few-shot learning and the symmetry prior

The original motivation for this analysis was data efficiency: children learn many concepts from a handful of examples; LLMs typically require enormous corpora. The order-of-magnitude story above explains *that there is a deep evolutionary prior*, but not *what the prior contains*. The KT corpus answers the second question.

Symmetries, not facts. In the Structured Dynamics paper (P7),[29] an agent's generative model is formalized as a smooth map $f : C \to \mathbb{R}^X$ from a low-dimensional configuration manifold C to high-dimensional observation space, with an associated Lie pseudogroup G acting on C. The world-tracking constraint — the agent's readout must match the current input — forces the agent's dynamics to be *equivariant* under G. The result is a Noether-style argument: tracking the data imposes a hierarchical organization on the agent's connectivity and dynamics that mirrors the symmetry properties of the generative world model. The agent's internal structure is, of necessity, the same shape as the world's invariance group.

The few-shot mechanism. Read alongside this paper, the picture is: what evolution has filtered for, over $\sim 10^{38}$–10^{40} cell-level events, is precisely the developmental program that builds nervous systems equivariant under the relevant world symmetries (continuity of objects, persistence of identity under translation and rotation, causal structure of bodies, social structure of conspecifics, ...). The child does not learn that objects persist under occlusion from a few examples; the inductive bias for object permanence is hardwired by the developmental program selected over evolutionary time. Few-shot generalization is then the data-efficient regime that necessarily holds when the agent's modeling engine already has the right invariance group in place.

Two corollaries follow.

Sample efficiency is depth, not magic. An AI system aspiring to child-level data efficiency must either (i) inherit a comparably deep prior — by pretraining on data that, after filtration, carries the right invariances (the implicit bet of large-scale self-supervised learning), or (ii) replace that depth with engineering: architecture, curriculum, scaffolding, dense supervision, or explicit inductive biases (the explicit bet of structured / equivariant / geometric deep learning). There is no third option; the empirical efficiency of biological cognition is paid for in evolutionary filtration depth.

The bridge. Evolution does not filter facts; it filters *generative priors* — developmental programs that build agents with useful invariances. A newborn organism is therefore not a blank model: it is the runtime instantiation of a hereditary program whose modeling, objective, and planning machinery already encode deep priors over the relevant invariance groups of its niche. Individual learning is fine-tuning on top of inherited structure. (Further KT-internal connections are deferred to a brief corpus-appendix.)

8 Conclusion

DNA is not the data. It is a compact hereditary program shaped by the filtered consequences of evolutionary history: produced by a vast, lossy, embodied, population-level filtration process. Evolution has far more exposure / evaluation events than LLM pretraining, but a far weaker and sparser write-back channel; LLMs learn from fewer exposures with dense gradient updates. The two converge to compressed survivor programs by very different mechanisms, on very different scales. The reason biological learners begin not from scratch, but from deep evolutionary structure, is this filtration depth: a structured prior of vast upstream search, encoded as the developmental program that builds an agent before individual experience begins.

A Relation to the KT corpus

This paper sits at the intersection of several existing BCOM working papers, each providing a piece the present quantitative framing depends on:

- **WP0049 (Coarse-Grained Computation)**[4] — the Wolpert–Korbel-style account of computation between dynamical systems: "A computes B iff $C(A) \cong C(B)$." Supplies the formal coarse-graining ladder used in §2.1.
- **WP0058 (Darwinian–Lamarckian Continuum)**[7] — the write-back vector $\boldsymbol{\lambda} = (\lambda_B, \lambda_F, \lambda_P)$ and the transmissible computational structure H_g. The present paper studies the $\boldsymbol{\lambda} \to \mathbf{0}$ regime.
- **WP0077 (Persistent Pattern)**[28] — persistence as mutual algorithmic information; the AIT definition of a "persistent pattern."
- **WP0056 (Liquid Brains and Agent Soups)** — collective lifetime computation across agents sharing high MAI; the source of paragraph (c) in §3.2.
- **WP0096 (Kolmogorov Manifesto)** — the informal statement that "DNA is a compressed representation of the environment," here made quantitative.
- **WP0110 (Genome to Exocortex)**[8] — the seven-stage substrate arc S1–S7. The cellular-lineage anchor of this paper is S1.
- **P7 (Structured Dynamics of the Algorithmic Agent)**[29] — the symmetry / Lie-pseudogroup argument for equivariance that underwrites the few-shot bridge in §7.
- **WP0018 (Mathematical Foundations of the Algorithmic Agent)**[6] — the formal Modeling Engine / Objective Function / Planning Engine triad in which a newborn organism is a runtime instantiation.

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